

Rohan Jaiswal

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SUMMARY

Graduate Computer Science student and Data Science Graduate Assistant with expertise in software engineering, data science, and AI/ML projects. Contributed to a YOLOv5-based vehicle recognition system with superior detection accuracy and enhanced data pipeline efficiency in large-scale analytical projects. Involved in crafting a thesis focused on scalable machine learning frameworks, underscoring strong capabilities in programming and cloud-based solutions. Eager to apply advanced analytical and technical skills to further engineering and research initiatives.

PROFESSIONAL EXPERIENCE

Texas Tech University | Data Science Graduate Assistant

Jun 2025 - Present

Supported a university-affiliated project with Nielsen, analyzing 1+ TB of retail data to uncover insights and inform strategic decisions.

- Leveraged Polars to boost analysis speed by 50% over pandas.
- Identified a 30% drop in seasonal demand, revealing reverse trends against expectations.
- Built memory-efficient ETL pipelines, reducing resource usage by 40%.
- Delivered dashboards and reports with 25% faster turnaround.
- Ensured data quality and alignment through cross-functional collaboration.

Texas Tech University | Data Science Research Assistant

Aug 2024 - Jun 2025

Developed a structured dataset (~198,000+ cells) to study the influence of AI technology acquisitions on organizational strategies across leading companies.

- Led a team of 5 with faculty supervision to design and annotate the dataset, completing the project ahead of schedule.
- Analyzed 750+ qualitative sources using QDA Miner and Atlas.ti, enabling structured evaluation across multiple case studies.
- Applied NLP and regression techniques in Python and R, uncovering 10+ key factors driving successful AI adoption.
- Created visualizations using Matplotlib, improving stakeholder interpretability and reducing early analysis turnaround time by 40%.
- Outcomes contributed to a strategic evaluation framework; dataset is being prepared for public release, and the research paper is currently under peer review.

Tata Consultancy Services | Assistant System Engineer

Jul 2022 - Jun 2023

- Developed client-specific software solutions in C++ using Agile methodology, leading to a 20% increase in client satisfaction and faster project delivery times.
- Worked on a YOLOv5-based vehicle recognition system, achieving 87% mAP on 5K+ images with enhanced detection accuracy.
- Analyzed marketing trends for enterprise clients using Pandas and Matplotlib to support data-driven decisions.
- Automated testing workflows and worked with cross-functional teams to increase product reliability and speed up releases by 20%.

PROJECTS

AI-Driven Quiz Application

Aug 2024 - Dec 2024

- Developed an AI-powered quiz generator by integrating Flask (Python), PHP, and the Gemini AI model.
- Automated quiz creation from lecture videos, reducing manual effort by over 80% and enabling real-time performance tracking for 50+ students.

Reinforcement Learning: Q-Learning & SARSA

Oct 2024

- Implemented Q-Learning and SARSA in a 10×10 maze environment for path optimization.
- Improved policy convergence near goal states by 40% through iterative value updates.
- Visualized learning behavior and state values using Matplotlib for performance analysis.

EDUCATION

Texas Tech University, Lubbock, TX, United States

Aug 2023 - Dec 2025

Master of Science, Computer Science

- **Coursework:** Thesis: A Reinforcement Learning Guided Hybrid Neural Framework for Sparse Feature Selection — Developing a hybrid method using reinforcement learning and attention mechanisms to improve feature selection in high-dimensional datasets.

TECHNICAL SKILLS

- **Programming/Databases:** C++, Python, Java, C, MySQL, SQLite, PostgreSQL, R Programming
- **Data Visualization:** Matplotlib, Seaborn, Tableau, PowerBI
- **Big Data & Cloud:** AWS, Apache Spark, PySpark, Kubernetes, Kafka, Distributed Systems
- **Data Science & AI:** Text Analysis, Reinforcement Learning, Data Modelling, Statistical Analysis, Machine Learning, Neural Networks, Regression Analysis, Time Series Analysis, A/B Testing, Natural Language Processing, Model Interpretability
- **Software & Development Tools:** GitHub, Excel, Anaconda, Visual Studio
- **Frameworks:** TensorFlow, PyTorch, Scikit-Learn, Keras, Django, Flask, OpenCV, NumPy, Pandas, PyQt